eLife

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 1
May 9, 2024 (this version)

Sent for peer review
February 21, 2024

Posted to preprint server
November 25, 2023

Neuroscience

The decoding of extensive samples of motor units in human muscles reveals the rate coding of entire motoneuron pools

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Abstract

To advance our understanding of the neural control of muscle, we decoded the firing activity of extensive samples of motor units in the Tibialis Anterior (129±44 per participant; n=8) and the Vastus Lateralis (130±63 per participant; n=8) during isometric contractions of up to 80% of maximal force. From this unique dataset, we characterised the rate coding of each motor unit as the relation between its instantaneous firing rate and the muscle force, with the assumption that the linear increase in isometric force reflects a proportional increase in the net synaptic excitatory inputs received by the motoneuron. This relation was characterised with a natural logarithm function that comprised two phases. The initial phase was marked by a steep acceleration of firing rate, which was greater for low-than medium- and high-threshold motor units. The second phase comprised a linear increase in firing rate, which was greater for high-than medium- and low-threshold motor units. Changes in firing rate were largely non-linear during the ramp-up and ramp-down phases of the task, but with significant prolonged firing activity only evident for medium-threshold motor units. Contrary to what is usually assumed, our results demonstrate that the firing rate of each motor unit can follow a large variety of trends with force across the pool. From a neural control perspective, these findings indicate how motor unit pools use gain control to transform inputs with limited bandwidths into an intended muscle force.

Significance statement

Movements are performed by motoneurons transforming synaptic input into an activation signal that controls muscle force. The control signal is not linearly related to the net synaptic input, but instead emerges from interactions between ionotropic and neuromodulatory inputs to motoneurons. Critically, these interactions vary across motoneuron pools and differ between muscles. We decoded the activity of hundreds of motor units from electromyographic signals recorded in vivo in humans to derive the most comprehensive framework to date of motor unit activity during isometric contractions that spanned the entire operating range of two human leg muscles. Our results address several key gaps in knowledge in this field and, importantly, characterise the activity of entire motor unit populations for each muscle.

eLife assessment

Leveraging state-of-the-art experimental and analytical approaches, this **valuable** study characterizes the recruitment and activation of large populations of human motor units during slow isometric contractions in two lower limb muscles. Evidence for many claims is **solid**, however, the main claim that this study reveals rate coding of entire motoneuron pools requires additional data in more dynamic conditions.

Introduction

Human muscles are versatile effectors producing forces that span several orders of magnitude. They allow humans to perform extremely diverse motor tasks with the same limbs, like a surgeon closing an incision, and a climber who grasps supports on a cliff. This versatility relies on a unique structure that converts neural inputs into muscle force; that is, the motor unit, which comprises an alpha motoneuron and the muscle fibres it innervates (1, 2). The nervous system controls muscle force by modulating the number of active motor units (recruitment) and their firing rates (rate coding) (3). This control strategy involves projecting a substantial level of common synaptic inputs to motoneurons (4-6) that recruits them in a fixed order following the size principle (7).

While we have a clear understanding of the general mechanisms of force control, we still lack a full picture of the detailed organisation of the firing activities of motor units. This is challenged by our inability, so far, to decode the concurrent firing activity of the full spectrum of motor units spanning the entire range of recruitment thresholds in human muscles (8). Indeed, much of our knowledge on the modulation of motor unit activity still derives from animal preparations (9-11) or from serial recordings of motor units in humans, mainly decoded over narrow force ranges (12-14).

Animal studies have shown two stages in the rate coding of individual motor units to increase muscle force (amplification and rate limiting) in response to linear increases in the net excitatory synaptic inputs (11, 15-17). These stages correspond to different responses of motoneurons to ionotropic inputs due to modulation of their intrinsic properties by metabotropic inputs acting on the somato-dendritic surfaces of motoneurons through several ion channels (2, 16, 17). Similar characteristics in firing activity have been observed in humans (12, 13, 18), although these observations have been limited to small samples of motor units (typically < 40) with firing activities decoded over a narrow range of submaximal forces (typically <30% maximal voluntary contraction (MVC) force) (12, 13, 18). It is therefore difficult to infer from these studies a control scheme that is generalisable to entire pool and to all contraction levels.

The combination of arrays of electromyographic (EMG) electrodes with modern source-separation algorithms set the path for the decoding of populations of active motor units - and their motoneurons - in humans (8, 19). Recent studies have identified the concurrent firing activity of up to 60 motor units per contraction and per muscle by increasing the density of electrodes (20, 21). Here, we further extended this approach by tracking motor units across contractions at target forces that ranged from 10 to 80 %MVC, based on the unique spatial distribution of motor unit action potentials across the array of surface electrodes (22, 23). We were able to identify up to ~200 unique active motor units per muscle and per participant in two human muscles *in* *vivo*, yielding extensive samples of motor units that are representative of the entire motoneuron pools (24). With this approach, we described the non-linear transformation of the net excitatory synaptic inputs into firing activity for individual motoneurons (2, 17). We specifically focused on the non-linearities between the changes in firing rate and force: amplification, saturation, and hysteresis (9, 17, 25). The results provided a framework picture of the rate coding of motor units during large increases and decreases in an applied force.

Having access to the rate coding of entire motoneuron pools allowed us to understand how ionotropic and neuromodulatory inputs combine to modulate force, confirming some details of hypothetical force-control schemes in humans. For example, experimental (26, 27) and computational (25) studies have proposed a gain-control mechanism driven by neuromodulatory inputs to motoneurons. One common conclusion of these studies is that pools of motor units involved in fine motor tasks have a low gain, whereas additional motor units activated during more forceful tasks exhibit a progressively increase in gain (17, 25-27). However, this conclusion is based on results from motor units in animals (16) and humans (26, 28, 29) while experimentally modulating inputs, or by fitting a non-linear model predicting muscle activation from excitatory synaptic inputs at multiple contraction levels (27). In contrast, we addressed this question by decoding the concurrent activity of hundreds of motor units spanning the entire range of recruitment thresholds in vivo in humans.

Results

Identification and tracking of individual motor units

16 participants performed either isometric dorsiflexion (n = 8) or knee extension tasks (n = 8) while we recorded the EMG activity of the tibialis anterior (TA - dorsiflexion) or the vastus lateralis (VL – knee extension), with four arrays of 64 surface electrodes (256 electrodes per muscle). The experimental tasks consisted of isometric contractions involving a ramp-up, a plateau, and a ramp-down phase. The ramp-up and ramp-down phases were relatively slow compared with contractile speeds observed during activities of daily living. They were performed at a constant pace of 5 %MVC.s^{-1} and the force plateau was maintained for either 10-s (70-80 %MVC), 15-s (50-60 %MVC) or 20-s (10-40 %MVC) (**Figure 1A**). The level of force during the plateau ranged between 10 and 80 %MVC, with increments of 10 %MVC (randomised order). A source-separation algorithm (8, 19) was applied to the EMG signals to extract motor unit pulse trains, from which we identified the series of discharge times. All identified motor unit firings were visually inspected and manually edited when necessary (30). Because the signal was stationary during the plateau, we were able to estimate reliable separation vectors for large samples of motor units with source-separation algorithms (8, 19) (**Figure 1A**). The average number of motor units identified in each contraction per participant was 42 ± 24 (25th–75th percentile: 24 – 53, up to 95) motor units from the TA and 33 ± 15 (25th–75th percentile: 23 – 47, up to 71) motor units from the VL. The datasets from all contraction levels were merged to obtain a sample of unique motor units per muscle and participant spanning the full range of recruitment thresholds (1st–99th percentile: 0.9-73.4% MVC; see below).

To estimate the proportion of the EMG signal represented by the identified motor units, a synthetic EMG signal was reconstructed from the firing activity. To do so, the discharge times were used as triggers to segment the differentiated EMG signals over a window of 25 ms and estimate the averaged action potential waveforms for each motor unit (**Figure 1A**). The action potentials were then convolved with the discharge times to obtain trains of action potentials, and all the trains of the identified motor units were summed to reconstruct the synthetic EMG signal. Finally, the ratio between the powers of synthetic and original EMG signals was calculated (**Figure 1B**).

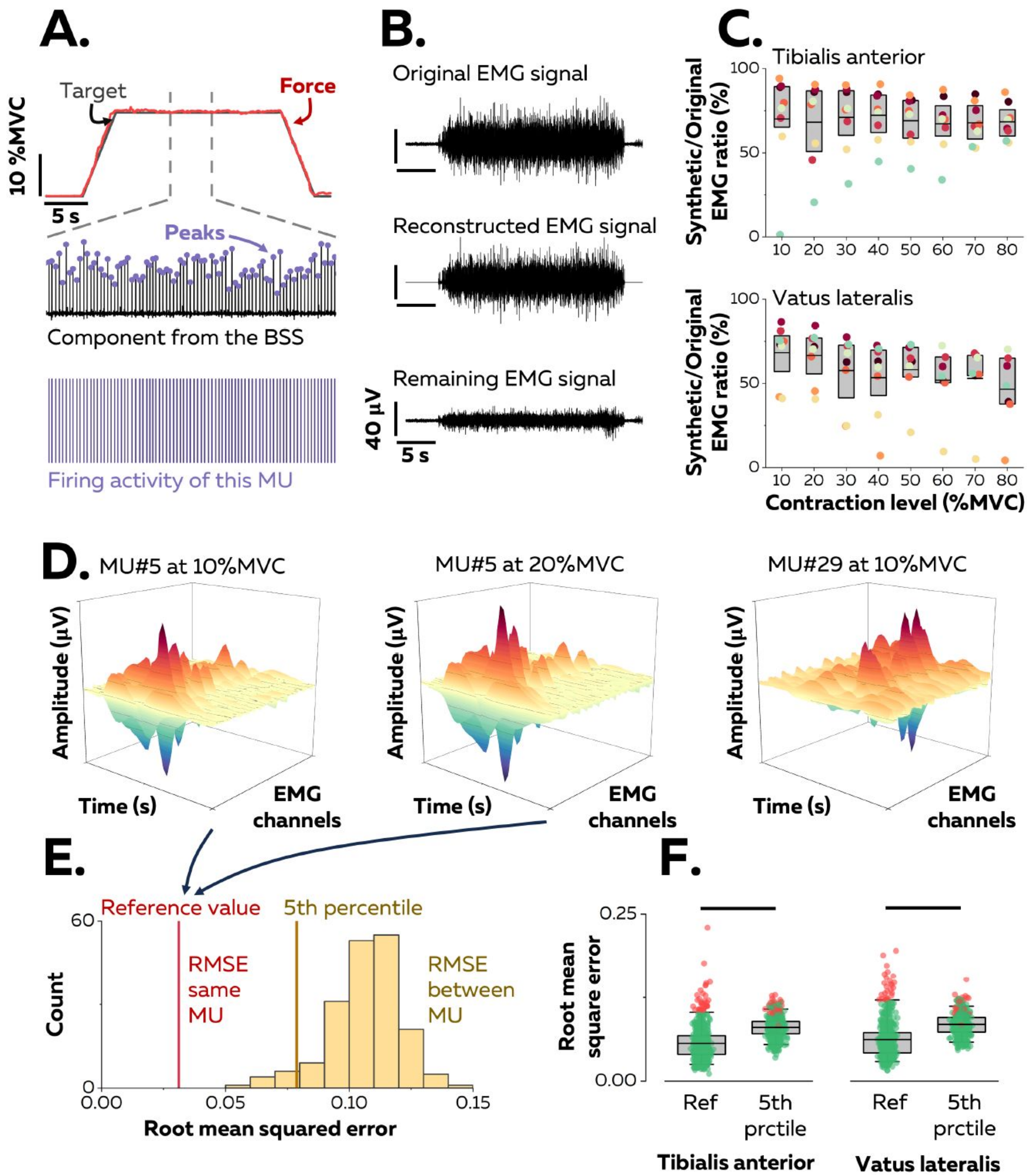

Figure 1.

Identification of extensive samples of motor units in two human muscles.

A. We used a blind source separation (BSS) algorithm to decompose the overlapping activity of motor units (MU) into trains of discharge times during a force-matching trapezoidal task. B. We reconstructed synthetic EMG signals by summing the trains of action potentials from all the identified motor units and interpreting the remaining EMG signal as the part of the signal not explained by the decomposition. C. We calculated the ratio between the powers of synthetic and original EMG signals to estimate the proportion of the signal variance explained by the decomposition. D. We estimated the uniqueness of each identified motor unit within the pool by calculating the root-mean-square error (RMSE) between the distributions of action potentials of the same motor unit across contractions (two panels on the left, reference value in E), and between motor units (left vs. right panels, distribution of RMSE between motor units in yellow in E). F. We considered each motor unit to be unique within the pool when the RMSE between its distributions of action potentials across contractions (reference value) was less than the 5th percentile of the distribution of RMSE with the rest of the motor units across contraction levels. Motor units considered as outliers in F (red data points) were removed from the analysis due to potential errors in tracking between contractions. Each data point is a motor unit, the box represents 25th-75th percentiles of the distribution of data, and the black line is the median. The horizontal thick line denotes a statistical difference between reference values and 5th percentiles for each muscle.

This ratio reached 69.3 ± 17.3% (25^{th}–75^{th} percentile: 59.3 – 83.6%, up to 94.2%) for TA and 55.2 ± 19.5% (25^{th}–75^{th} percentile: 50.0 – 71.9%, up to 86.5%) for VL (**Figure 1C**), meaning that most of the recorded EMG signals were successfully decomposed into motor unit activity.

To investigate motor unit activity across large force ranges, motor units were tracked between contractions using their unique spatial distribution of action potentials (23) (**Figure 1D**). This method was validated beforehand using both simulated EMG signals and two-source validation with simultaneous recordings of firing activity from intramuscular and surface EMG signals (Figure S1). On average, we tracked 67.1 ± 10.0% (25^{th}–75^{th} percentile: 53.9 – 80.1%) of the motor units between consecutive contraction levels (10% increments, e.g., between 10% and 20% MVC) for TA and 57.2 ± 5.1% (25^{th}–75^{th} percentile: 46.6 – 68.3%) of the motor units for VL (Figure S2). There are two explanations for the inability to track all motor units across consecutive contraction levels. First, some motor units are recruited at higher targets only. Second, it is challenging to track small motor units beyond a few contraction levels due to signal cancellation (31, 32). In total, we identified 129 ± 44 motor units (25^{th}–75^{th} percentile: 100 – 164, up to 194) per participant in TA and 130 ± 63 motor units (25^{th}–75^{th} percentile: 103 – 173, up to 199) per participant in VL.

The accuracy of the tracking method was further tested by confirming the uniqueness of each motor unit within the identified sample, assuming that each motor unit must have a unique representation of their action potentials across the array of surface electrodes (22). This was accomplished by calculating the root mean-square error (RMSE) between their action potentials across the electrodes and the action potentials of the rest of the motor units (**Figure 1E**). The RMSE between the action potentials of the same motor unit tracked across contractions was calculated as a reference and compared with the 5^{th} percentile of the distribution of RMSE between motor units (**Figure 1F**). This reference value typically less than the 5^{th} percentile of the RMSE calculated with the action potentials of the other motor units (TA: 5.6 ± 2.4 vs. 8.0 ± 1.5%; $p<0.001$; VL: 6.2 ± 2.8% vs. 8.5 ± 1.6%; $p<0.001$). Inspection of data for each motor unit revealed that 92.1% (TA) and 87.0% (VL) of the motor units had a lower RMSE between contractions than with the other motor units (<5^{th} percentile). Of note, motor units with the highest reference values (>95^{th} percentile) were considered as outliers due to tracking errors and were excluded from the subsequent analyses (**Figure 1F**). As a result of this approach, a total of 34 motor units were excluded from TA and 28 from VL.

Non-linear rate coding during the linear increase in force – amplification and rate limiting

The input-output function for each motoneuron was characterised as the relation between its instantaneous firing rate and the muscle force (**Figure 2A**). The linear increase in isometric force was assumed to reflect a proportional increase in the net synaptic excitatory inputs received by the motoneurons, as proposed previously (13, 33). Thus, any deviation from a linear increase in firing rate with respect to a linear increase in force should reflect the influence of neuromodulatory inputs on motoneuron gain (13, 14, 18), or a saturation of the motoneuron firing rate (12, 13). Therefore, the association between the firing rate and the force was characterised by comparing three curve fits: i) a linear fit, which would indicate an increase in firing rate proportional to the increase in the net synaptic excitatory input, ii) a rising exponential fit, which would correspond to an initial acceleration of firing rates followed by full saturation (12, 13, 34), and iii) a natural logarithm fit, which would denote an initial acceleration of firing rate followed by a slower constant increase in firing rate that reflects a rate limiting effect (11, 15, 35) (**Figure 2A**). Of note, this analysis was performed on each unique motor unit with the data pooled across all contractions (**Figure 2A**). The result was that 69.5% of the motor units in TA and 72.3% of those in VL had an input-output function better fitted by a natural logarithm (**Figure 2B**, inset panels).

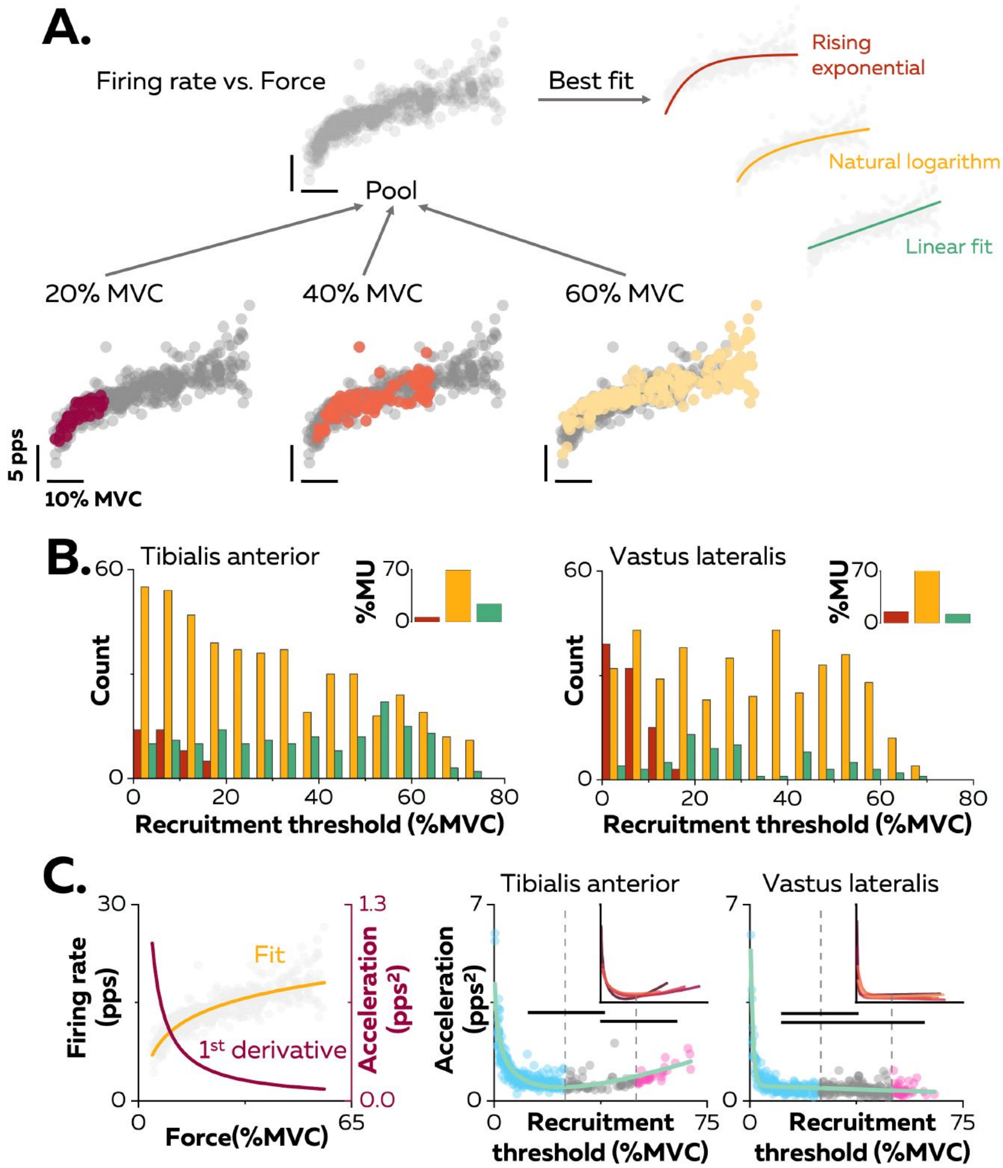

Figure 2.

Non-linear rate coding of motor units.

A. We analysed the relation between motor unit firing rate (pulses per second, pps) and force with the assumption that a linear increase in force is proportional to a linear increase in net synaptic input. For this, we concatenated the values of instantaneous firing rate for each motor unit (grey data points) recorded over all the contractions where that motor unit was identified, as shown here for one motor unit (coloured data points for contractions at 20, 40, and 60%MVC). The force-firing rate relations were then fitted with three different functions: linear (green), rising exponential (dark red), and natural logarithm (yellow), to characterise the input-output function of each motoneuron. B. The motor units were grouped according to their best fit. We display the distribution of motor unit recruitment thresholds (RT) for each of these groups. The inset panels depict the percentage of motor units (MU) in each group. C. We further analysed the rate coding of motor units by reporting the acceleration of firing rate. For this, we fitted the force-firing rate relation with the natural logarithm (f(force) =a*ln(force)+b; yellow trace) and computed its first derivative (f(force)=a/force; dark red trace). The right panels depict the relations between the initial acceleration of motor unit firing rates and their recruitment threshold (RT) for all participants, with a total of 328 and 393 motor units for TA and VL, respectively. Each data point is a motor unit. The horizontal thick line denotes a statistical difference between motor units grouped according to their recruitment thresholds (low = Blue; medium = Grey; high = Pink). The green line depicts the non-linear fits of these relations for the Tibialis Anterior and the Vastus Lateralis. Similar fits were observed for all the participants (inset panels).

The input-output functions of the motor units were then compared based on the distribution of recruitment thresholds (1^{st}–99^{th} percentile: 0.9-73.4% MVC). We clustered motor units into three groups with equal ranges of recruitment thresholds: low- (0 to 25% MVC), medium- (25 to 50% MVC), and high- (50 to 75% MVC) threshold motor units (**Figure 2B**). A significant percentage of input-output functions of low threshold motor units was better fitted with a rising exponential function (12.5% for the TA, 30.9% for the VL), highlighting a steep acceleration of firing rate followed by a full saturation. This pattern was not observed in medium- and high-threshold motor units (0% for TA and VL). In contrast, a significant percentage of input-output functions of high threshold motoneurons was better fitted with a linear function for TA (39.6%), although the input-output functions of high threshold motor units from VL were still better fitted with natural logarithms (87.9%) than linear functions (12.1%).

We further described the first stage of rate coding by estimating the acceleration of firing rate (**Figure 2C**; Figure S3). The initial acceleration of the firing rate, likely due to the amplification of synaptic inputs by persistent inward currents (11), was calculated as the value of the first derivative of the natural logarithm function at the recruitment threshold (**Figure 2C**). These values were compared between motor units using a linear mixed effect model, with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect. There was a significant effect of *Muscle* (F=30.4; p<0.001), *Threshold* (F=24.7; p<0.001), and a significant interaction *Muscle* × *Threshold* (F=4.1; p=0.017). The initial acceleration of firing rate in TA was greater for the low- (1.0±0.7 pps^2; p<0.001) and the high- (1.0±0.3 pps^2; p=0.032) threshold motor units than for medium-threshold motor units (0.6±0.3 pps^2), with no difference between the low- and high-threshold motor units (p=0.999) (**Figure 2C**). Similarly, the initial acceleration of firing rate was greater for low-threshold motor units (0.6±0.6 pps^2) than for medium- (0.4±0.2 pps^2; p<0.001) and the high-threshold motor units (0.4±0.2 pps^2; p=0.013) in VL, with no difference between the medium- and high-threshold motor units (p=0.990) (**Figure 2C**). The initial acceleration of firing rate was also greater for the low- and high-threshold motor units from TA comparing with VL (p<0.001 for both).

Finally, we tested whether the distribution of initial accelerations of firing rate observed across the entire motor unit pool was generalisable to all the participants. Participants with either less than 20 motor units or with no motor units recruited below 5%MVC were excluded from this analysis (TA: n=4; VL: n=3). The function representing the relation between recruitment thresholds and initial accelerations were accurately fitted using the non-linear least-square fitting method (TA; adjusted R-squared: 0.77; VL; adjusted R-squared: 0.80; green lines on **Figure 2C**). We then fitted the same non-linear functions for each participant; the adjusted R-squared were increased to 0.86 ± 0.04 for the TA and 0.84 ± 0.11 for the VL (inset panels in **Figure 2C**).

The second stage of rate coding involved a linear increase in firing rate after the initial acceleration phase. This phase was analysed using the average firing rate during the successive force plateaus for each tracked motor unit (**Figure 3**). Specifically, the increase in firing rate between consecutive contraction levels (i.e., change in force equal to 10% MVC) was compared with a linear mixed effect model with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect. There were significant effects for *Muscle* (F=39.7; p<0.001) and *Threshold* (F=145.3; p<0.001), and a significant interaction *Muscle* × *Threshold* (F=37.7; p<0.001). The rate of increase was significantly greater for TA (3.5 ± 1.7 pps; 25^{th}-75^{th} percentile: 2.4 – 4.1 pps) than for VL (2.0 ± 1.1 pps; 25^{th}-75^{th} percentile: 1.3 – 2.6 pps. p<0.001). The rate of increase was significantly greater in TA for high-threshold than for low- (p<0.001) and medium- (p<0.001) threshold motor units and for medium-than for low-threshold motor units (p<0.001). In contrast, the rate of increase in VL was greater for high- (p<0.001) and medium- (p<0.001) threshold motor units than for low-threshold motor units, with no difference between high- and medium-threshold motor units in VL (p=0.993).

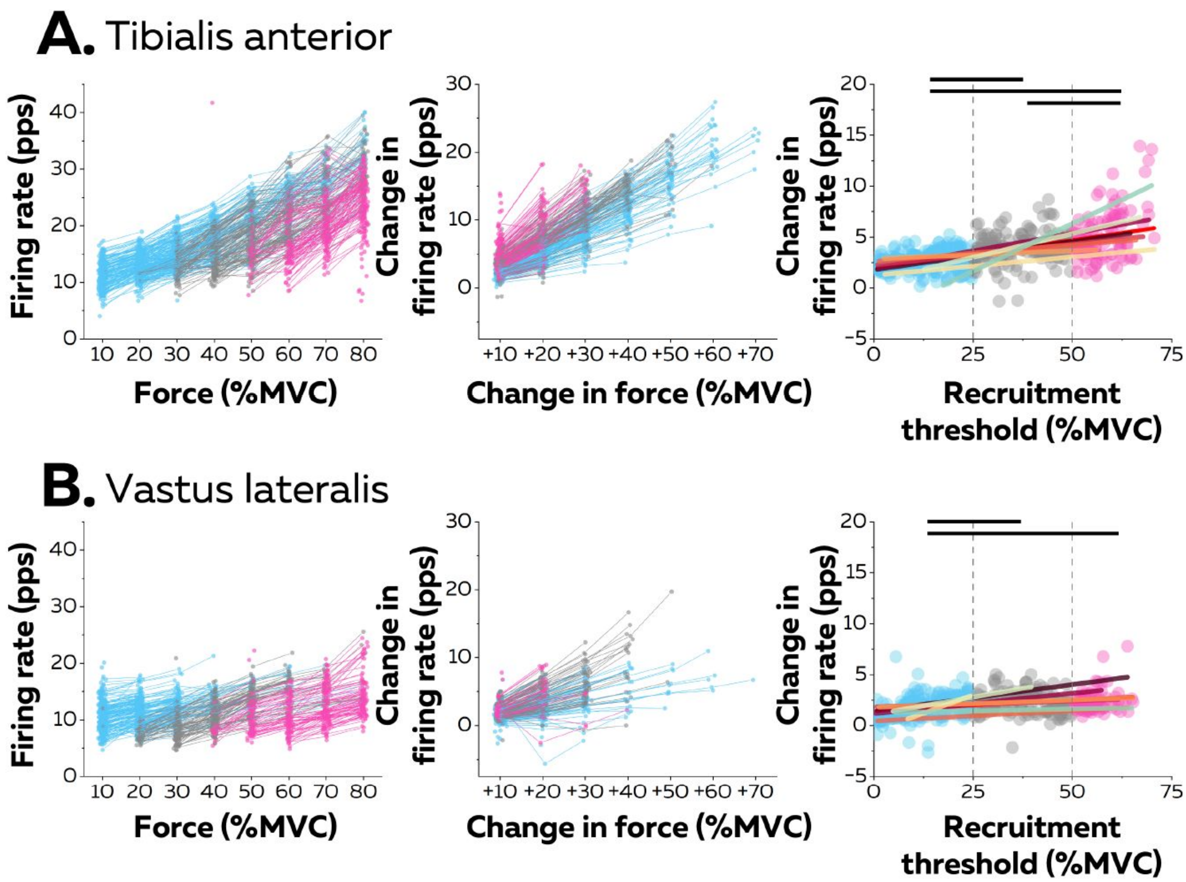

Figure 3.

Motor unit firing rates across contraction levels.

We calculated the average firing rate (pulses per second, pps) during the force plateaus for each tracked motor unit for all participants from TA (n = 998 motor units; A) and VL (n = 1016 motor units; B). Each data point is a motor unit, and each line connects the firing rates of this motor unit across contractions. The colour scale identifies the three groups of motor units based on recruitment threshold: low (blue), medium (grey), and high (pink). The central panels depict the change in firing rates between contractions separated by 10% to 70% of the maximal voluntary contraction level (MVC). The right panels show the relation between the rate of increase in firing rate between successive levels of force (e.g., between 10% and 20%MVC) and the recruitment threshold of the motor unit. We fitted these relations for each participant with a linear function (coloured lines). The horizontal thick line denotes a statistical difference between motor units grouped according to their recruitment thresholds.

The data for each participant was characterised with a linear function (**Figure 3C**). There was a significant positive association between the rate of increase in firing rate and the recruitment threshold for all the participants (Pearson's r: TA: 0.71±0.14 pps and VL:0.57±0.22 pps; all p<0.022), except in one participant for each muscle (TA: r=0.17; p=0.272; VL: r= 0.12; p=0.411). Most motor units (92.0% in TA and 80.9% in VL) increased firing rates by more than one pulse per second between contractions despite reaching the maximal force tested or until tracking was not possible.

Differences between ascending and descending levels of force - hysteresis

Neuromodulatory inputs can prolong excitatory synaptic inputs and produce a hysteresis between the recruitment and the derecruitment thresholds. The left columns in **Figure 4** show the relation between recruitment and derecruitment thresholds within the motor unit pools (**Figure 4**). The relation for most participants was better characterised with a non-linear than a linear regression (TA: 6 out of 8 participants, adjusted R-squared >0.92); VL: 7 out of 8 participants, adjusted R-squared >0.83), indicating potential differences in the level of hysteresis between motor units of different size. The difference between the recruitement and the derecruitment threshold was compared with a linear mixed effect model with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect. There was a significant effect of *Threshold* (F=83.0; p<0.001) and a significant interaction *Muscle* × *Threshold* (F=30.1; p<0.001), but no effect of *Muscle* (F=2.5; p<0.137). We further tested for each group of motor units whether the difference between the two thresholds was significantly different from 0 (absence of hysteresis). Thus, only medium-threshold motor units had a significant positive hysteresis in both muscles (TA: −3.8 ± 7.2 pps; p = 0.013; VL: −3.1 ± 6.0 pps; p = 0.026), whereas low-threshold motor units from both muscles (TA: −1.5 ± 4.8; VL: −1.0 ± 3.2 pps; p>0.221) did not exhibit significant hysteresis. Although high-threshold motor units from TA followed the same trend (−1.7 ± 6.0 pps; p<0.437), high-threshold motor units from VL exhibited a negative hysteresis (VL: +3.2 ± 6.7 pps; p=0.039), that indicated that derecruitment threshold was greater than the recruitment threshold.

We also compared the relation between motor unit firing rate and force between the ramp-up and ramp-down phases of the isometric contraction. Most motor units (TA: 60.2%; VL: 76.7%) kept the same relation during the two phases. Although the percentage of associations better fitted with linear functions increased for the ramp-down phase for the TA (from 25.4% to 41.9% of motor units), there was no change between ramps for the VL (from 12.8% to 19.4% of motor units). Thus, most motor units kept a non-linear relation during the ramp-down phase.

Discussion

No previous studies have decoded to date the firing activity of >100 motor units spanning the full spectrum of recruitment thresholds, whether in animals or humans. According to previous estimates, the number of motor units in human would be 200±61 for TA and 146±29 for VL (36). We have therefore identified in this study most of the active motor units for each muscle (**Figure 1**), which has allowed us to infer the rate coding of entire pools of motor units. Overall, we found that motor units within a pool exhibit distinct rate coding with changes in force level (**Figure 2** and **3**), which contrasts with the long-held belief that rate coding is similar across motor units from the same pool.

The fast initial acceleration phase reflects the amplification of synaptic inputs through the activation of persistent inward currents via voltage-gated sodium and calcium channels (2, 11, 15, 17). Although noradrenergic and serotonergic synapses that generate persistent inward currents are widespread within motor unit pools (37), the influence of the

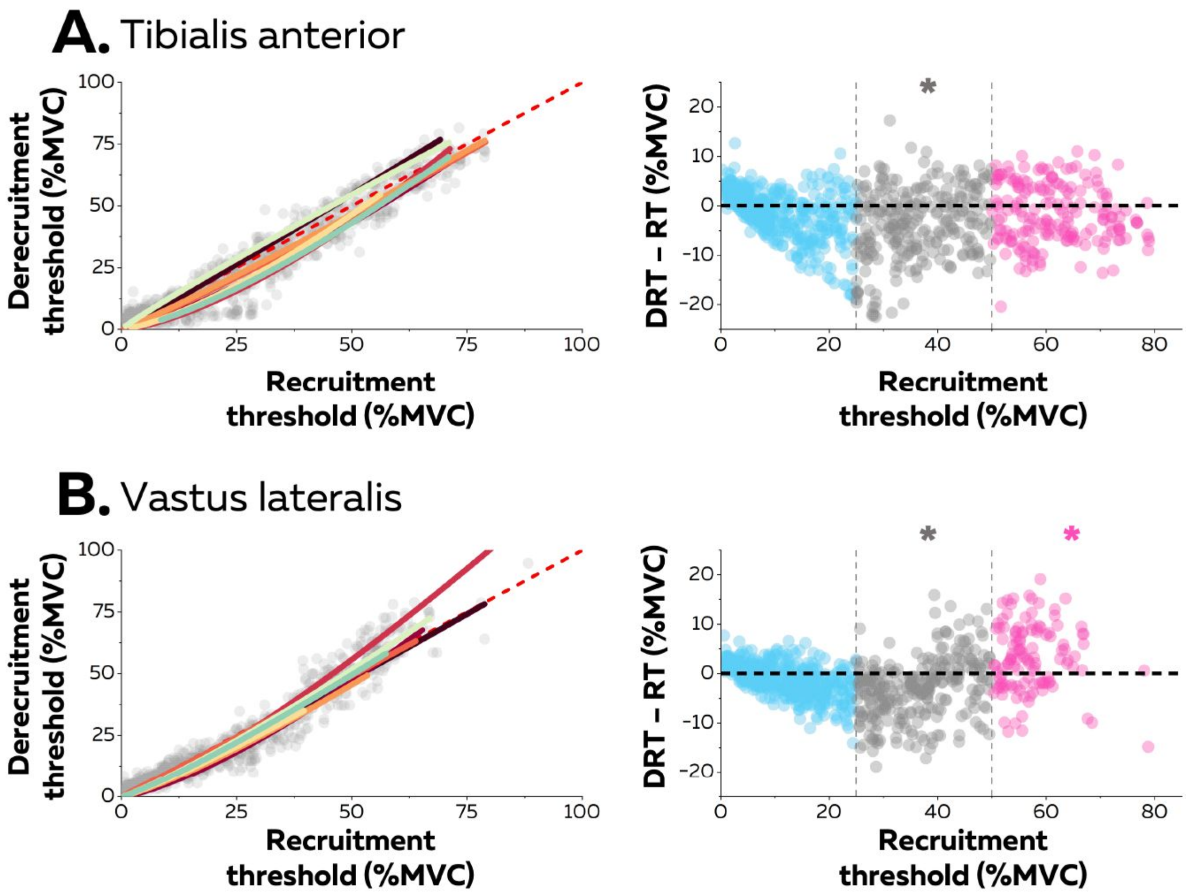

Figure 4.

Hysteresis between recruitment and derecruitment thresholds.

The left panels depict the relations between the recruitment and derecruitment thresholds of each motor unit from TA (A) and VL (B) across all participants. Each grey data point is a motor unit. These relations were fitted for each participant (coloured lines) using either non-linear or linear regressions. The values below the dashed red line (recruitment threshold = derecruitment threshold) show a positive hysteresis between recruitment and derecruitment thresholds, the values above a negative hysteresis. The right panels show the difference between the recruitment and derecruitment thresholds, with negative values showing a positive hysteresis with recruitment threshold greater than derecruitment threshold and the positive values indicating the converse (a negative hysteresis). Each data point is a motor unit. The asterisk denotes a statistical difference between the hysteresis values for motor units grouped according to recruitment threshold (low = Blue; medium = Grey; high = Pink) and the absence of a hysteresis (dashed horizontal line).

neuromodulatory inputs is greater during this phase for low-than for high-threshold motor units. This is presumably due to a larger ratio between the amplitude of inward currents and the input conductance in low threshold motoneurons (38).

The second phase of rate coding involves a slower linear increase in firing rate, as characterised by the second part of a logarithmic function (**Figures 2**-**3**). This phase corresponds to mechanisms previously referred to as 'rate limiting' (2, 25, 39) or 'saturation' (12, 34), and was more pronounced for low-than high-threshold motor units (**Figure 3**). This difference may be explained by smaller excitatory synaptic inputs onto low-than high-threshold motoneurons (2, 16), lower synaptic driving potential of the dendritic membrane (12, 40, 41), and longer and larger afterhyperpolarisation phase in low-than high-threshold motoneurons (42-45).

Taken together, these results show how ionotropic and neuromodulatory inputs to motoneurons uniquely combine to generate distinct rate coding across the pool, following a similar pattern across the pool, but with a pace that depends on motor unit size. While the size of the motor unit determines its recruitment threshold, neuromodulatory inputs determine its gain. Thus, in a similar fashion as size imposes a fixed recruitment order (7), neuromodulatory inputs determines a spectrum of variations in motor unit firing rates without the need to differentiate the ionotropic inputs to each motoneuron. (**Figure 2C-3**-S3) (25, 46, 47). Because high-threshold motor units exhibit a higher gain than low-threshold motor units (**Figure 3**), their firing rate could eventually reach similar or even greater values than that of low-threshold motor units during strong contractions (47-50). This indicates that the onion skin principle (34, 51) may not hold in all muscles and for all contraction forces (47-50).

It is also worth noting that the profile of rate coding mostly followed the same natural logarithmic trend during the ramp-up and ramp-down phases of the contraction, but with prolonged firing activity only evident for medium-threshold motor units (**Figure 4**). This hysteresis is possible because of the bistability of the membrane potential mediated by inward currents from calcium channels, the two stable membrane states being the resting potential and a depolarised state that enables self-sustained firing (9, 52). The absence of statistically significant hysteresis in low threshold motoneurons may be simply explained by their early recruitment during the first percentages of force, mathematically lowering the potential amplitude of their hysteresis. Alternatively, prominent outward currents may attenuate the impact of persistent inward currents on the prolongation of their firing activity (53). Similarly, the absence of hysteresis (TA) and even a negative hysteresis (VL) in high-threshold motoneurons can be explained by a briefer membrane bistability due to a faster decay of inward currents from calcium channels, and a narrower range of membrane depolarisation over which these inward currents are activated (52).

The increase in firing rate was also significantly greater for TA motor units than for those in VL. This difference may reflect a varying balance between excitatory/inhibitory synaptic inputs and neuromodulation (14, 25, 39, 46, 47, 54). Specifically, the strength of recurrent and reciprocal inhibitory inputs to motoneurons innervating VL and TA, and their proportional or inverse covariation with excitatory inputs, respectively, may explain the differences in rate limiting and maximal firing rates (14, 25, 39, 46, 47, 54). It is worth noting that similar differences in rate coding were previously observed between proximal and distal muscles of the upper limb (55).

Our results on rate coding characteristics of the entire motor unit pool provide insight on force control strategies. The accurate control of force depends on the firing activity of the population of active motor units, which effectively filters out the synaptic noise of individual motor units and primarily transforms common synaptic inputs into muscle force (4, 5, 56). Because the bandwidth of muscle force is <10 Hz during isometric contractions (57), the activation of inward currents to low-threshold motoneurons at recruitment may enable them to promptly discharge at a rate (>8 pps) that transmits the effective common synaptic inputs (<8-10 Hz) without phase distortion (4, 5, 56). This mechanism facilitates the accuracy of force control.

Moreover, force generation can be described as the filtering of the cumulative firing activity of active motor units with the average twitch force of their muscle units (5, 58). From this perspective, at low forces, recruitment of new motor units has a higher impact on force modulation than rate coding since each recruitment impacts both the cumulative firing activity of the pool and the average twitch force. Rate coding only modulates the cumulative firing activity. Thus, the amplification of the firing rate of low-threshold motor units near their recruitment threshold instantly favours rate coding over the recruitment of additional motor units, which likely allows for smoother force control. Similarly, the progressive recruitment of motor units with a higher gain promotes faster changes in firing rates, which promotes force control accuracy across the full force range. Overall, our results may help to design future non-linear decoders that aim to predict muscle activation or muscle force from descending inputs recorded in the motor cortex, with the will to generalise their performance across movements (27).

Methods

Detailed materials and methods are provided in supplementary files. Data and code are provided at *https://github.com/simonavrillon/MUdictionnary*.

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Reviewer #1 (Public Review):

Summary:

This study explores the neural control of muscle by decomposing the firing activity of constituent motor units from the grid of surface electromyography (EMG) in the Tibialis (TA) Anterior and Vastus Lateralis (VL) during isometric contractions. The study involves extensive samples of motor units across the broadest range of voluntary contraction intensities up to 80% of MVC. The authors examine the rate coding of the population of motor units, which describes the instantaneous firing rate of each motor unit as a function of muscle force. This relationship is characterized by a natural logarithm function that delineates two distinct phases: an initial phase with a steep acceleration in firing rate, particularly pronounced in low-threshold motor units, and a subsequent modest linear increase in firing rate, more significant in high-threshold motor units.

Strengths:

The study makes a significant contribution to the field of neuromuscular physiology by providing a detailed analysis of motor unit behavior during muscle contractions in a few ways.

(1) The significance lies in its comprehensive framework of motor unit activity during isometric contractions in a broad range of intensities, providing insights into the non-linear relationship between the firing rate and the muscle force. The extensive sample of motor units across the pool confirms the observation in animal studies in which the spinal motoneuron exhibits a discharge consisting of distinct phases in response to synaptic

currents, under the influence of persistent inward currents. As such, it is now reasonable to state the human motor units across the pool are also under the control of gain modulation via some neuromodulatory effects in addition to synaptic inputs arising from ionotropic effects.

(2) The firing scheme across the entire motoneuron pool revealed in this study reconciles the discrepancy in firing organization under debate; i.e., whether it is 'onion skin' like or not (Heckman and Enoka 2012). The onion skin like model states that the low threshold motor units discharge higher than high threshold motor units and have been held for a long time because the firing behaviors were examined in a partial range of contraction force range due to technical limitations. This reconciliation is crucial because it is fundamental to modelling the organization of motor unit recruitment and rate coding to achieve a desired force generation to advance our understanding of motor control.

(3) The extensive data collection with a novel blind source separation algorithm on the expanded number of channels of surface EMG signal provides a robust dataset that enhances the reliability and validity of findings, setting a new standard for empirical studies in the field.

Collectively, this study fills several knowledge gaps in the field and advances our understanding of the mechanism underlying the isometric force generation.

Weaknesses:

Although the findings and claims based on them are mostly well aligned, some accounts of the methods and claims need to be clarified.

(1) The authors examine the input-output function of a motor unit by constructing models, using force as an input and discharge rate as an output. It sounds circular, or the other way around to use the muscle force as an input variable, because the muscle force is the result of motor unit discharges, not the cause that elicits the discharges. More specifically, as a result of non-linear interactions of synchronous and/or asynchronous discharges of a population of a given motoneuron pool that give rise to transient increase/maintenance in twitch force, the gross muscle force is attained. I acknowledge that it is extremely challenging experimentally to measure synaptic currents impinging upon the spinal motoneurons in human subjects and the author has an assumption that the force could be used as a proxy of synaptic currents. However, it is necessary to explicitly provide the caveats and rationale behind that. Force could be used as the input variable for modelling.

1. The authors examine the firing organizations in TA and VL in this study without explicit purposes and rationale for choosing these muscles. The lack of accounts makes it hard for the readers to interpret the data presented, particularly in terms of comparing the results from the different muscles.

(3) In the methods, the author described the manual curation process after applying the blind source separation algorithm. For the readers to understand the whole process of decomposition and to secure rigor and robustness of the analyses, it would be necessary to provide details on what exact curation is performed with what criteria.

(4) In Figure 3, the early recruited units tend to become untraceable in the higher range of contraction. This is more pronounced in the muscle VL. This limitation would ambiguate the whole firing curve along the force axis and therefore limitation and the applicability in the different muscles needs to be discussed.

(5) It is unclear how commonly the notion "the long-held belief that rate coding is similar across motor units from the same pool" is held among the community without a reference. Different firing organizations have been modelled and discussed in the seminal paper by

Fuglevand et al. (1993), and as far as I understand, the debate has not converged to a specific consensus. As such, any reference would be required to support the claim the notion is widely recognized.

(6) The authors claim that the firing behavior as a function of force is well characterized by a natural logarithmic function, which consists of initial steep acceleration followed by a modest increase in firing rate. Arguably the gain modulation in firing rate could be attributed to a neuromodulatory effect on the spinal motoneuron, which has been suggested by a number of animal studies. However, the complexity of the interactions between ionotropic and neuromodulatory inputs to motoneurons may require further elucidation to fully understand the mechanisms of neural control; it is possible to consider the differential acceleration among different threshold motor units as a differential combinatory effect of ionotropic and neuromodulatory inputs, but it is not trivially determined how differentially or systematically the inputs are organized. Likewise, the authors make an account for the difference in firing rate between TA and VL in terms of different amounts or balances of excitatory and inhibitory inputs to the motoneuron pool, but again this could be explained by other factors, such as a different extent of neuromodulatory effects. To determine the complexity of the interactions, further studies will be warranted.

(7) It is unclear with the account " ... the bandwidth of muscle force is < 10Hz during isometric contraction" in the manuscript alone, and therefore, it is difficult to understand the following claim. It appears very interesting and crucial for motor unit discharge and force generation and maintenance because it would pose a question of why the discharge rate of most motor units is higher than 10Hz, despite the bandwidth being so limited, but needs to be elaborated.

(References)

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https://doi.org/10.7554/eLife.97085.1.sa2

Reviewer #2 (Public Review):

Summary:

The motivation for this study is to provide a comprehensive assessment of motor unit firing rate responses of entire pools during isometric contractions. The authors have used new quantitative methods to extract more unique motor units across contractions than prior studies. This was achieved by recording muscle fibre action potentials from four high-density surface electromyogram (HDsEMG) arrays (Caillet et al., 2023), quantifying residual EMG comparing the recorded and data-based simulation (Figure 1A-B), and developing a metric to compare the spatial identification for each motor unit (Figure 1D-E). From identified motor units, the authors have provided a detailed characterization of recruitment and firing rate responses during slow voluntary isometric contractions in the vastus lateralis and tibialis anterior muscles up to 80% of maximum intensity. In the lower limb, it is interesting how lower threshold motor units have firing rate responses that saturate, whereas higher threshold units that presumably produce higher muscle contractile forces continue to increase their firing rate. In many ways, these results agree with the rate coding of motor units in the extensor digitorum communis muscle (Monster and Chan, 1977). The paper is detailed, and the analyses are well explained. However, there are several points that I think should be addressed to strengthen the paper.

General comments:

(1) The authors claim they have measured the complete rate coding profiles of motor units in the vastus lateralis and tibialis anterior muscles. However, this study quantified rate coding during slow and prolonged voluntary isometric contractions whereas the function of rate coding during movements (Grimby and Hannerz, 1977) or more complex isometric contractions (Cutsem and Duchateau, 2005; Marshall et al., 2022) remains unexplored. For example, supraspinal inputs may not scale the same way across low and higher threshold motor units, or between muscles (Devanne et al., 1997), making the response of firing rates to increasing isometric contraction force less clear. Conceptually, the authors focus on the literature on intrinsic motoneurone properties, but in vivo, other possibilities are that descending supraspinal drive, spinal network dynamics, and afferent inputs have different effects across motor unit sizes, muscles, and types of contractions. Also, the influence from local muscles that act as synergists (e.g., vastii muscles for the vastus lateralis, and peroneal muscles that evert the foot for the tibialis anterior) or antagonists (coactivation during higher contraction intensities would stiffen the joint) may provide differential forms of proprioceptive feedback across motor pools.

(2) The evidence that the entire motor unit pool was recorded per muscle is not clear. There appears to be substantial residual EMG (Figure 1B), signal cancellation of smaller motor units (lines 172-176), some participants had fewer than 20 identified motor units, and contractions never went above 80% of MVC. Also, to my understanding, there remains no gold-standard in awake humans to estimate the total motor unit number in order to determine if the entire pool was decomposed. Furthermore, using four HDsEMG arrays also raises questions about how some channels were placed over non-target muscles, and if motor units were decomposed from surrounding synergists.

(3) The authors claim (Abstract L51; Discussion L376) that a commonly held view in the field is that rate coding is similar across motor units from the same pool. Perhaps this is in reference to some studies that have carefully assessed lower threshold motor units during lower force ramp contractions (e.g., Fuglevand et al., 2015; Revill and Fuglevand, 2017). However, a more complete integration of the literature exploring motor unit firing rate responses during rapid isometric contractions, comparing different muscles and contraction intensities would be helpful. From Figure 3, the range of rate coding in the tibialis anterior (~7-40 Hz) is greater than the vastus lateralis (~5-22 Hz) muscle across contraction levels. In agreement with other studies, the range of rate coding within some muscles is different than others (Kirk et al., 2021) and during maximal intensity (Bellemare et al., 1983) or rapid contractions (Desmedt and Godaux, 1978). Likewise, within a motor pool, there is a diversity of firing rate responses across motor units of different sizes as a function of isometric force (Monster and Chan, 1977; Desmedt and Godaux, 1977; Kukula and Clamann, 1981; Del Vecchio et al., 2019; Marshall et al., 2022). A strength of this paper is how firing rate responses are quantified across a wide range of motor unit recruitment thresholds and between two muscles. I suggest improving clarity for the general reader, especially in the motivation for testing two lower limb muscles, and elaborating on some of the functional implications.

https://doi.org/10.7554/eLife.97085.1.sa1

Reviewer #3 (Public Review):

Summary:

This is an interesting manuscript that uses state-of-the-art experimental and simulation approaches to quantify motor unit discharge patterns in the human TA and VL. The non-linear profiles of motor unit discharge were calculated and found to have an initial acceleration phase followed by an attenuation phase. Lower threshold motor units had a larger gain of the initial acceleration whereas the higher threshold motor unit had a higher gain in the attenuation phase. These data represent a technical feat and are important for understanding how humans generate and control voluntary force.

Strengths:
The authors used rigorous, state-of-the-art analyses to decompose and validate their motor unit data during a wide range of voluntary efforts.

The analyses are clearly presented, applied, and visualized.

The supplemental data provides important transparency.

Weaknesses:

The number of participants and muscles tested are quite small - particularly given the constraints on yield. It is unclear if this will translate to other motor pools. The justification for TA and VL should be provided.

While an impressive effort was made to identify and track motor units across a range of contractions, it appears that a substantial portion of muscle force was not identified. Though high-intensity contractions are challenging to decompose - the authors are commended for their technical ability to record population motor unit discharge times with recruitment thresholds up to 75% of a participant's maximal voluntary contractions. However previous groups have seen substantial recruitment of motor units above 80% and even 90% maximum activation in the soleus. Given the innervation ratios of higher threshold motor units, if recruitment continued to 100%, the top quartile would likely represent a substantial portion of the traditional fast-fatigable motor units. It would be highly interesting to understand the recruitment and rate coding of the highest threshold motor units, at a minimum I would suggest using terms other than "entire range" or "full spectrum of recruitment thresholds"

The quantification of hysteresis using torque appears to make self-evident the observation that lower threshold motor units demonstrate less hysteresis with respect to torque. If there is motor unit discharge there will be force. I believe this limitation goes beyond the floor effects discussed in the manuscript. Traditionally, individuals have used the discharge of a lower threshold unit as the measure on which to apply hysteresis analyses to infer ion channel function in human spinal motoneurons.

The main findings are not entirely novel. See Monster and Chan 1977 and Kanosue et al 1979.

https://doi.org/10.7554/eLife.97085.1.sa0